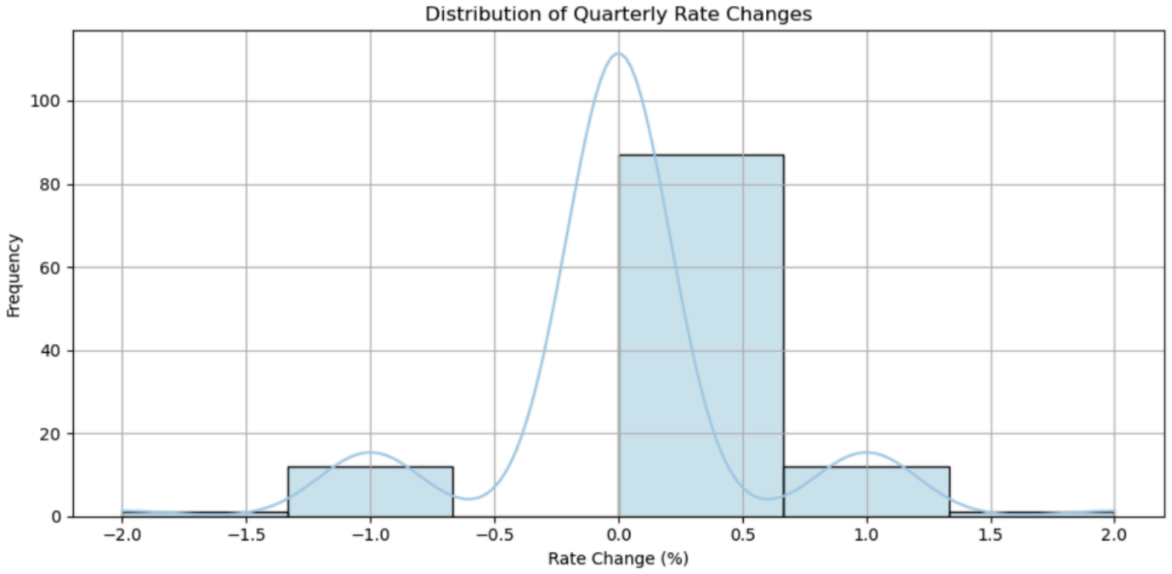
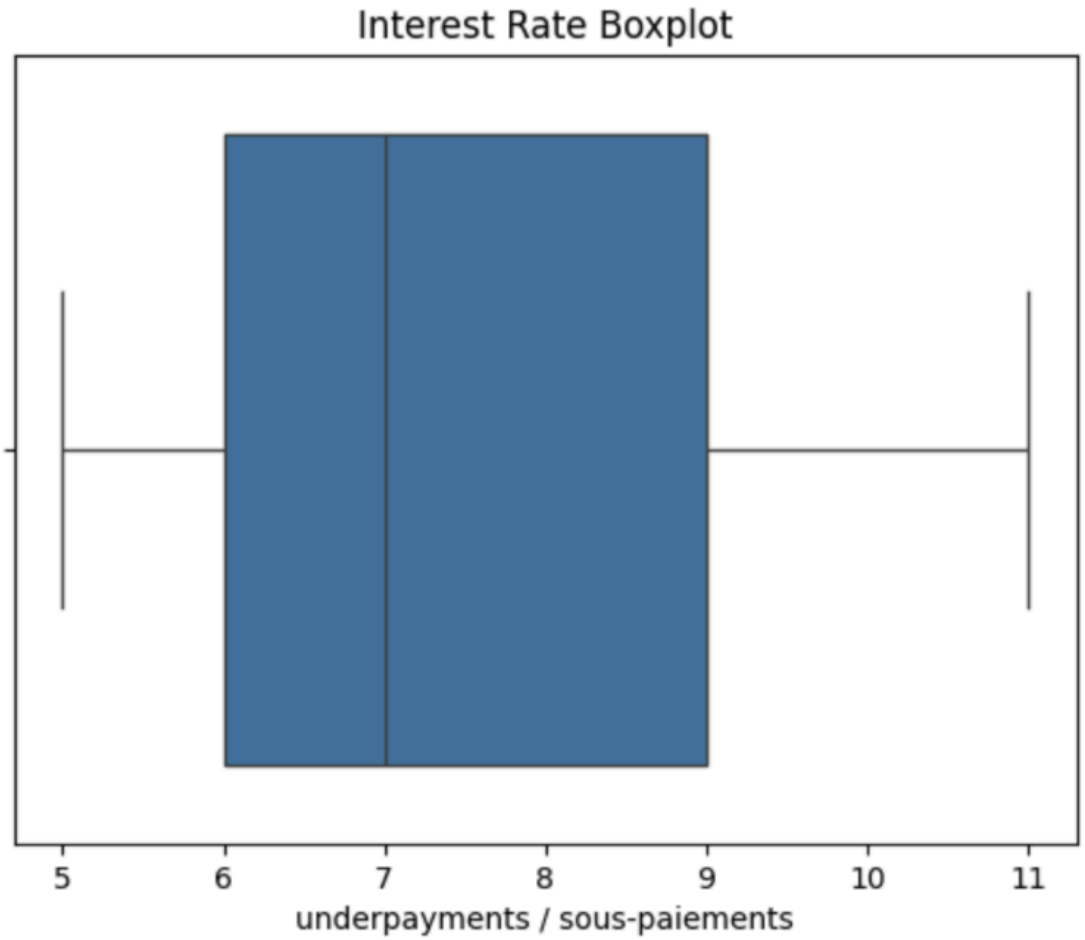
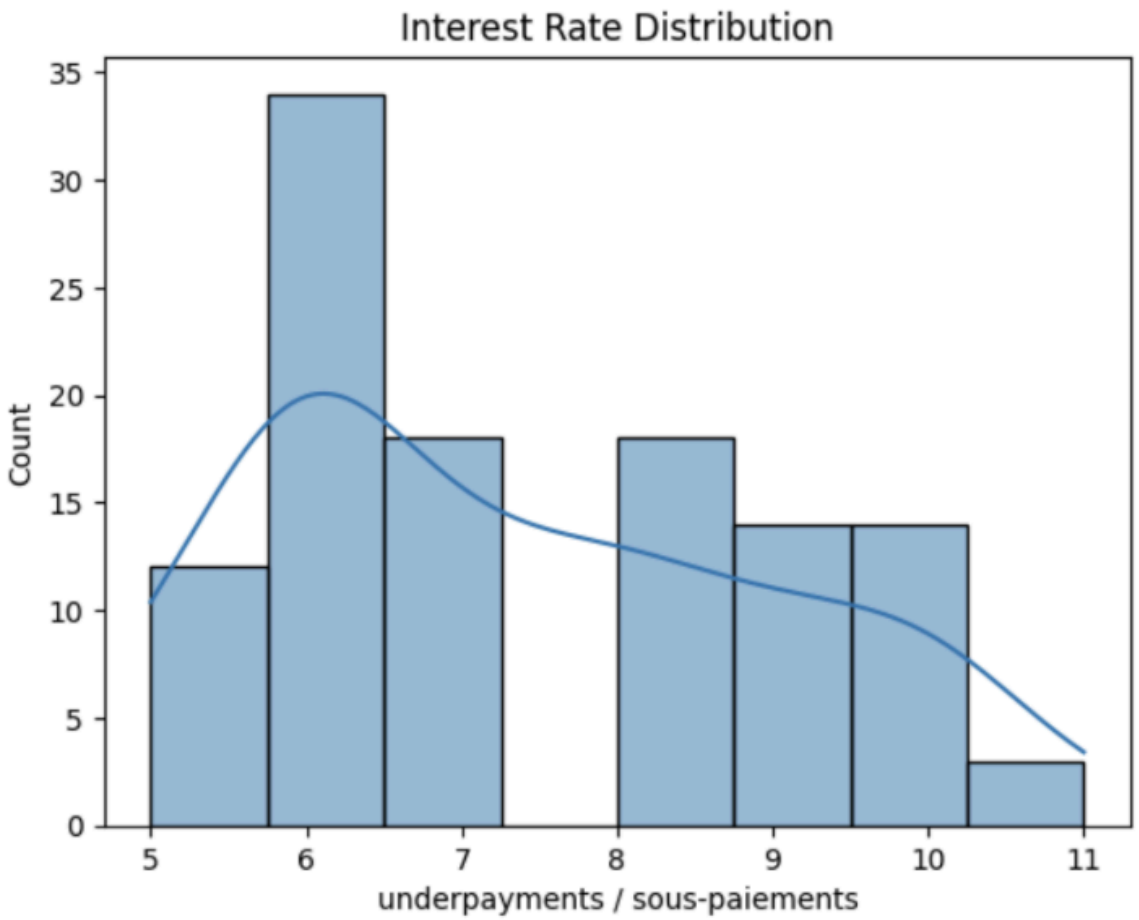
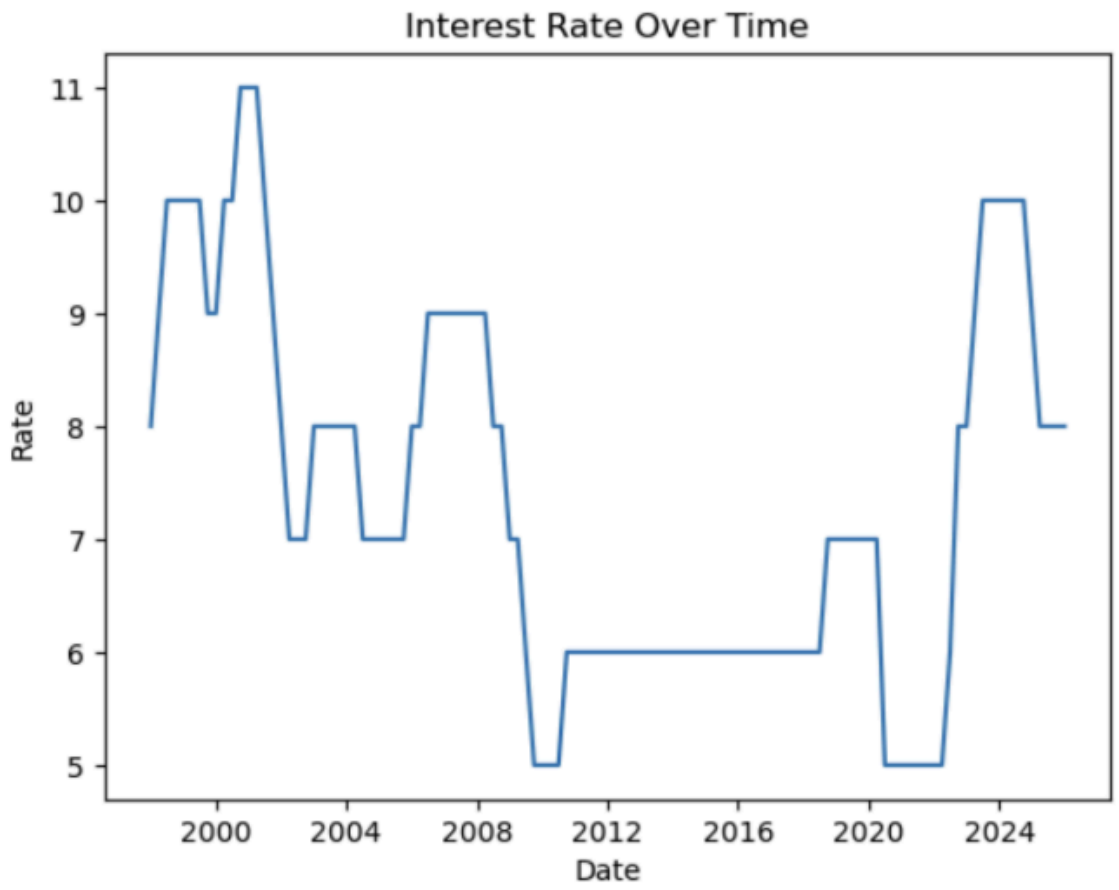
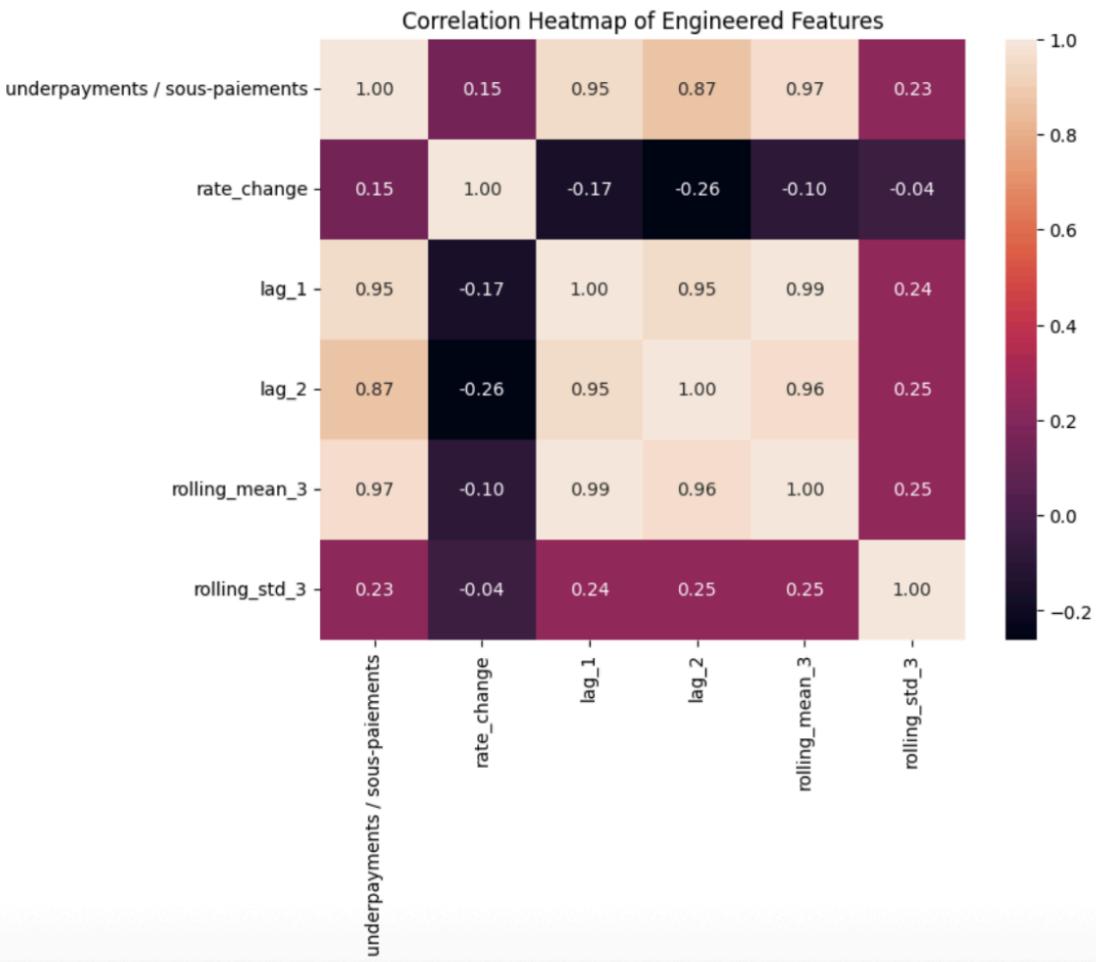
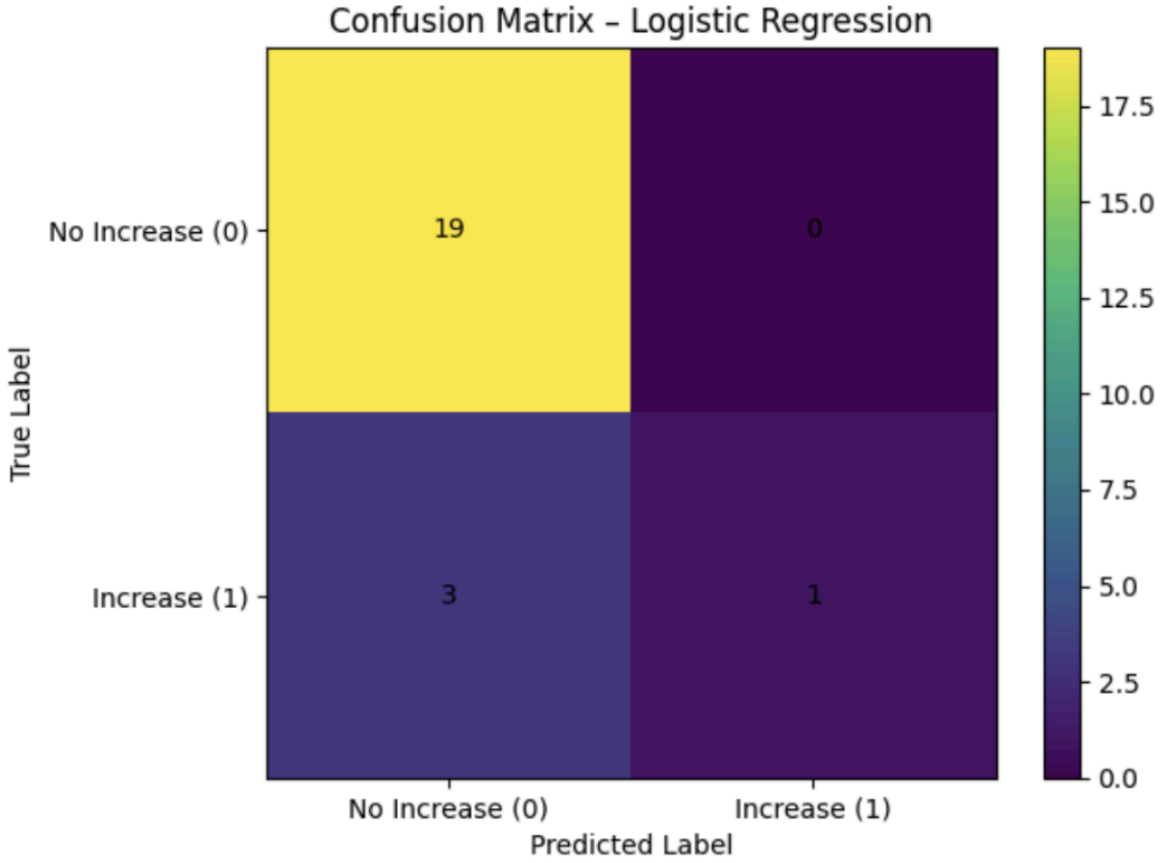

Distribution of Quarterly Rate Changes
Frequency
Rate Change (%)
0
20
40
60
80
100
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0

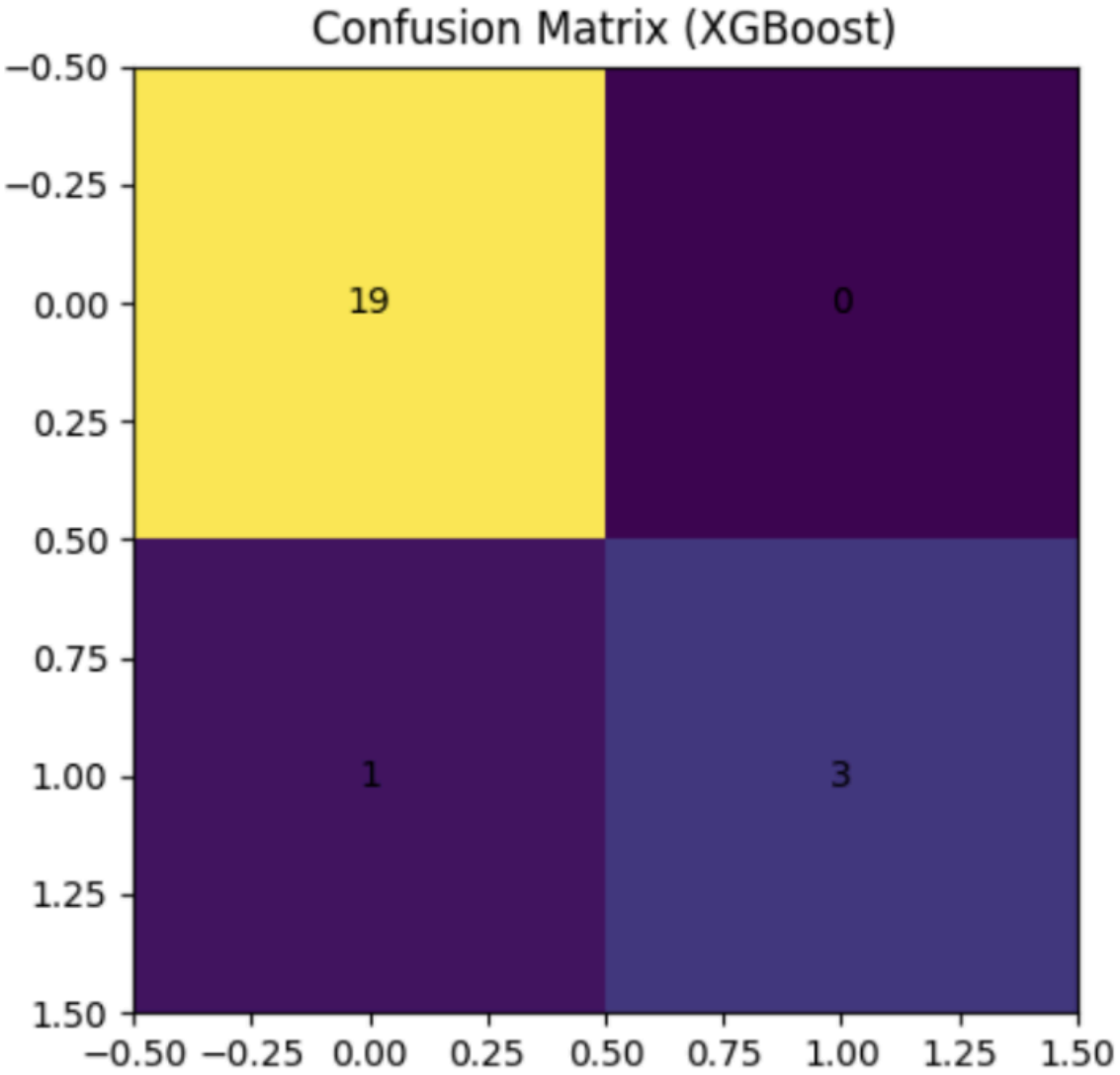

Interest Rate Boxplot
5
6
7
8
9
10
11
underpayments / sous-paiements

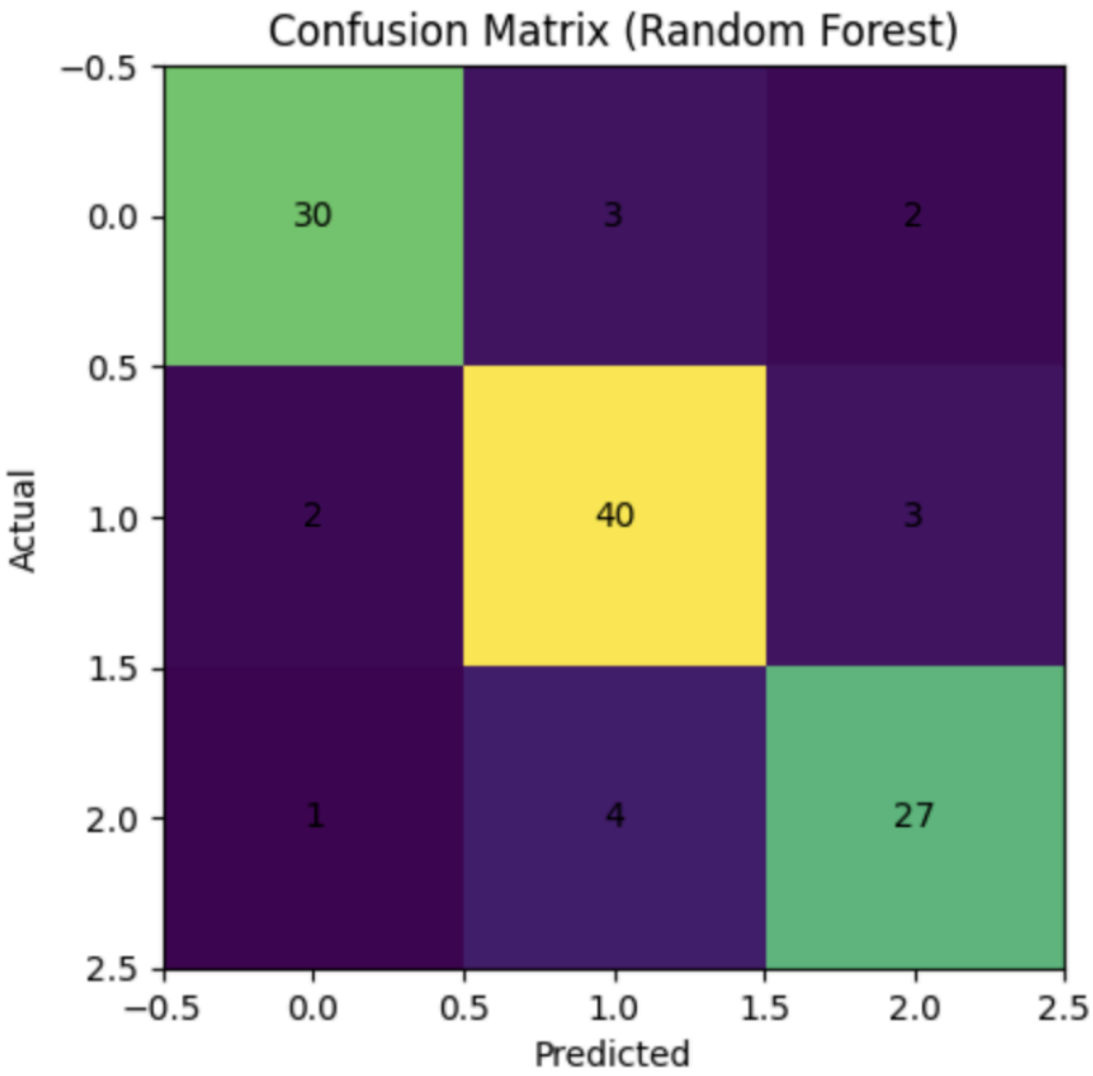

Interest Rate Distribution
Count
35
30
25
20
15
10
5
0
5
6
7
8
9
10
11
underpayments / sous-paiements

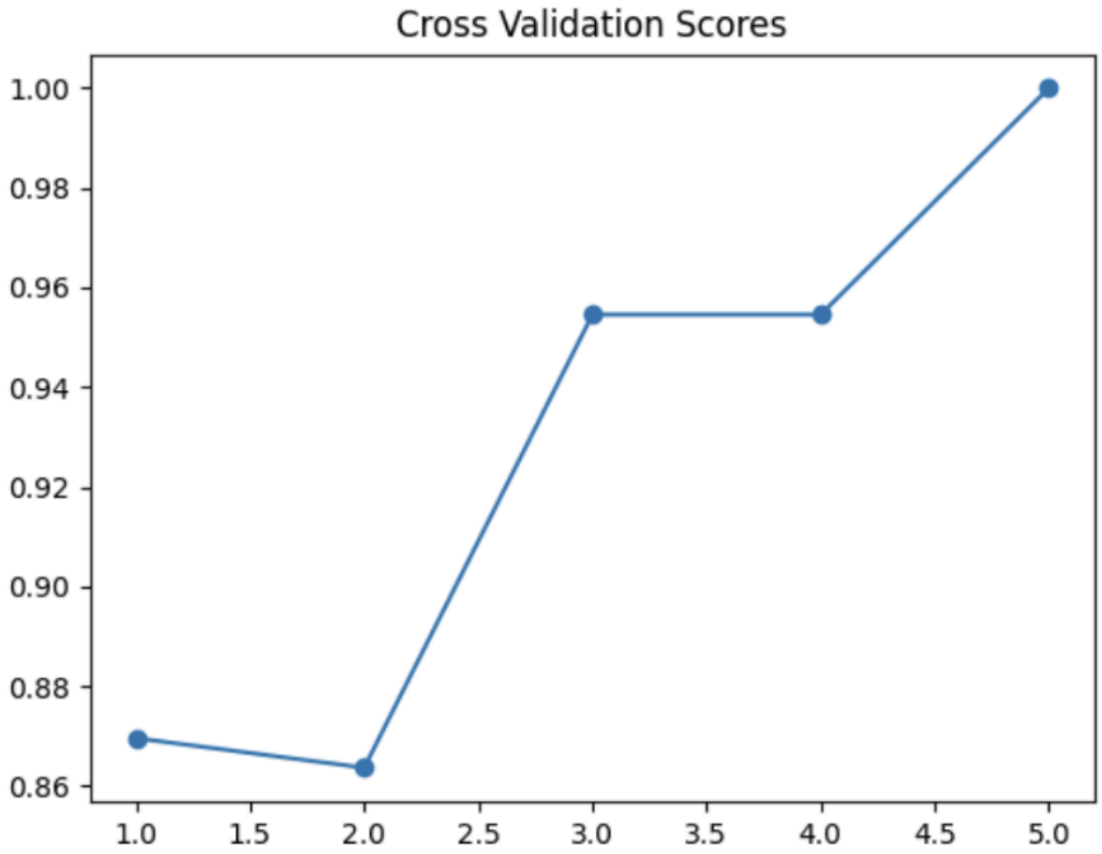

Interest Rate Over Time
Rate
Date
11
10
9
8
7
6
5
2000
2004
2008
2012
2016
2020
2024

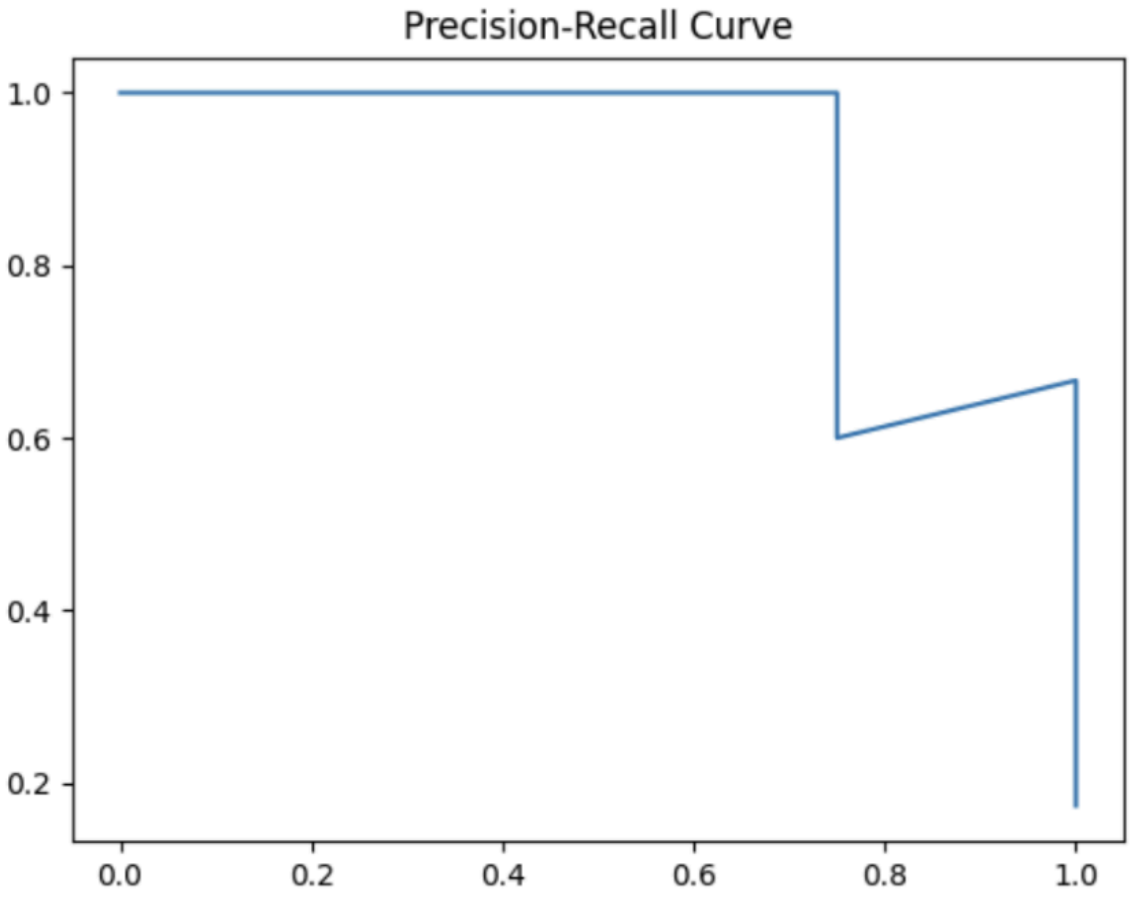

Correlation Heatmap of Engineered Features
underpayments / sous-paiements
rate_change
lag_1
lag_2
rolling_mean_3
rolling_std_3
1.00 0.15 0.95 0.87 0.97 0.23
0.15 1.00 -0.17 -0.26 -0.10 -0.04
0.95 -0.17 1.00 0.95 0.99 0.24
0.87 -0.26 0.95 1.00 0.96 0.25
0.97 -0.10 0.99 0.96 1.00 0.25
0.23 -0.04 0.24 0.25 0.25 1.00
1.0
0.8
0.6
0.4
0.2
0.0
−0.2


Confusion Matrix – Logistic Regression
No Increase (0)
Increase (1)
True Label
19
0
3
1
No Increase (0)
Increase (1)
Predicted Label
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0


Confusion Matrix (XGBoost)
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
1.25
1.50
19
0
1
3
−0.50 −0.25 0.00 0.25 0.50 0.75 1.00 1.25 1.50


Confusion Matrix (Random Forest)
Actual
Predicted
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
30
3
2
2
40
3
1
4
27


Cross Validation Scores
1.00
0.98
0.96
0.94
0.92
0.90
0.88
0.86
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0


Precision-Recall Curve
1.0
0.8
0.6
0.4
0.2
0.0
0.2
0.4
0.6
0.8
1.0

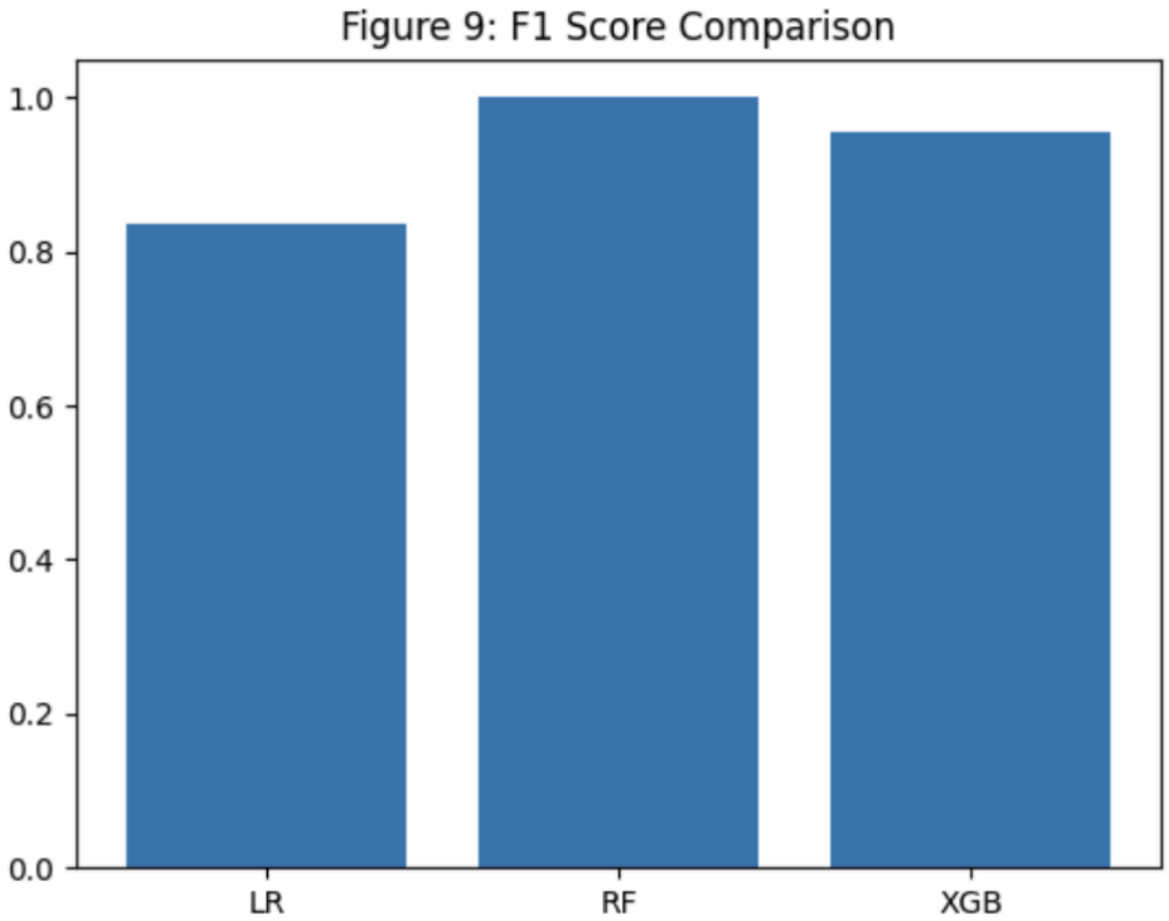


Figure 9: F1 Score Comparison
1.0
0.8
0.6
0.4
0.2
0.0
LR
RF
XGB

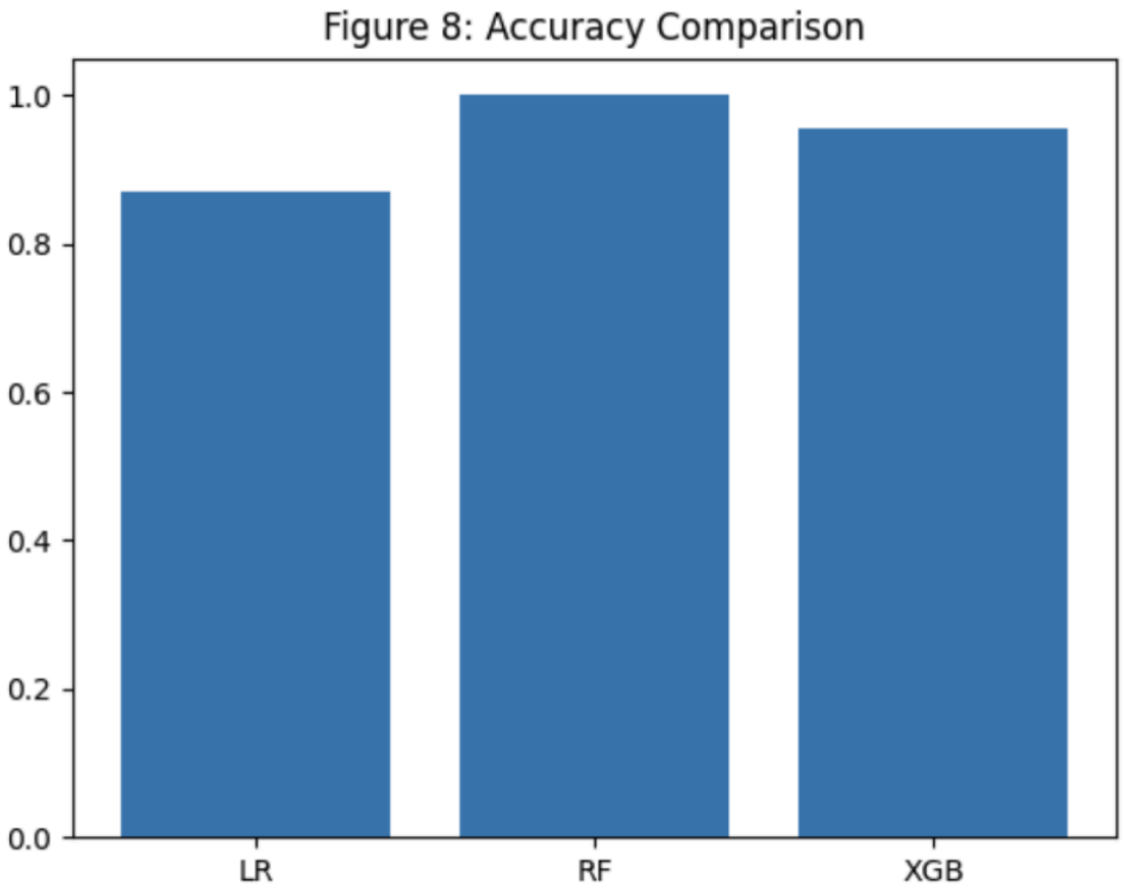


Figure 8: Accuracy Comparison
1.0
0.8
0.6
0.4
0.2
0.0
LR
RF
XGB